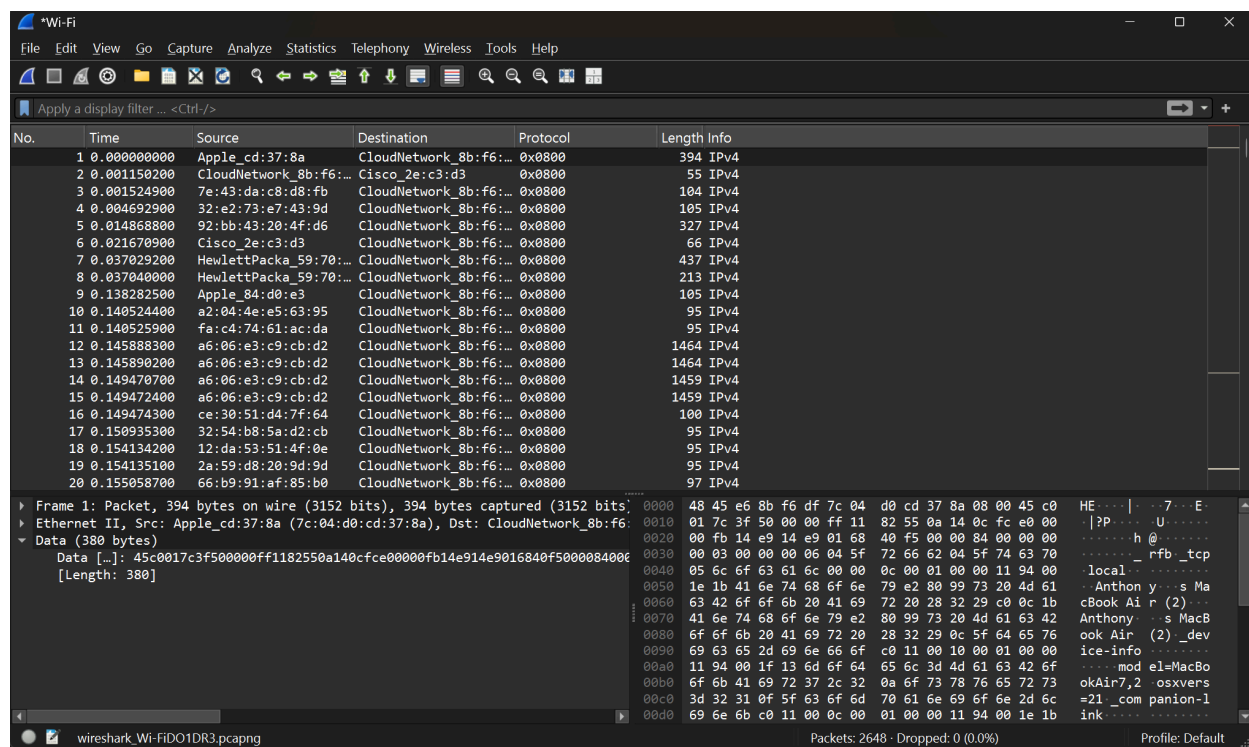


Ethernet and ARP Lab



After unchecking ipv4

We want to find corresponding MAC for

```
952  2.442760800  10.20.11.119  128.119.245.12  HTTP  549  GET
/wireshark-labs/HTTP-ethereal-lab-file3.html HTTP/1.1
```

Ethernet and ARP Lab

The image shows a Wireshark packet capture window. The top pane displays a list of captured packets. Packet 952 is selected, and the bottom pane shows its details. The Ethernet II section is expanded, showing the source and destination MAC addresses. The IP section is also expanded, showing the source and destination IP addresses. The packet bytes pane shows the raw data of the packet, with the Ethernet header (14 bytes) and IP header (20 bytes) highlighted in red.

No.	Time	Source	Destination	Protocol	Length	Info
934	2.308021200	HewlettPacka_59:70:...	CloudNetwork_8b:f6:...	0x0800	66	IPv4
935	2.308022500	Cisco_2e:c3:d3	CloudNetwork_8b:f6:...	0x0800	225	IPv4
936	2.308373200	CloudNetwork_8b:f6:...	Cisco_2e:c3:d3	0x0800	66	IPv4
937	2.308450900	CloudNetwork_8b:f6:...	Cisco_2e:c3:d3	0x0800	66	IPv4
938	2.308479000	CloudNetwork_8b:f6:...	Cisco_2e:c3:d3	0x0800	66	IPv4
939	2.308505100	CloudNetwork_8b:f6:...	Cisco_2e:c3:d3	0x0800	66	IPv4
940	2.308567800	CloudNetwork_8b:f6:...	Cisco_2e:c3:d3	0x0800	66	IPv4
941	2.309327200	CloudNetwork_8b:f6:...	Cisco_2e:c3:d3	0x0800	85	IPv4
942	2.309455400	CloudNetwork_8b:f6:...	Cisco_2e:c3:d3	0x0800	54	IPv4
943	2.309842300	CloudNetwork_8b:f6:...	Cisco_2e:c3:d3	0x0800	66	IPv4
944	2.309924200	CloudNetwork_8b:f6:...	Cisco_2e:c3:d3	0x0800	54	IPv4
945	2.355669900	CloudNetwork_8b:f6:...	Cisco_2e:c3:d3	0x0800	54	IPv4
946	2.356043100	Intel_c0:8d:69	IPv4mcast_7f:ff:fa	0x0800	143	IPv4
947	2.361285200	CloudNetwork_8b:f6:...	Cisco_2e:c3:d3	0x0800	66	IPv4
948	2.442138000	Cisco_2e:c3:d3	CloudNetwork_8b:f6:...	0x0800	66	IPv4
949	2.442138900	Cisco_2e:c3:d3	CloudNetwork_8b:f6:...	0x0800	66	IPv4
950	2.442292700	CloudNetwork_8b:f6:...	Cisco_2e:c3:d3	0x0800	54	IPv4
951	2.442342500	CloudNetwork_8b:f6:...	Cisco_2e:c3:d3	0x0800	54	IPv4
952	2.442760800	CloudNetwork_8b:f6:...	Cisco_2e:c3:d3	0x0800	549	IPv4
953	2.448391500	Cisco_2e:c3:d3	CloudNetwork_8b:f6:...	0x0800	933	IPv4

Frame 952: Packet, 549 bytes on wire (4392 bits), 549 bytes captured (4392 bits) on interface 0

Ethernet II, Src: CloudNetwork_8b:f6:df (48:45:e6:8b:f6:df), Dst: Cisco_2e:c3:d3 (9c:09:8b:2e:c3:d3)

Source: CloudNetwork_8b:f6:df (48:45:e6:8b:f6:df)

Type: IPv4 (0x0800)

[Stream index: 1]

Data (535 bytes)

Data [..]: 45000217055d4000800600000a140b778077f50ccd7200505395ec85ba1ba1875 [Length: 535]

0000 9c 09 8b 2e c3 d3 48 45 e6 8b f6 df 08 00 45 00HE.....E

0010 02 17 05 5d 40 00 80 06 00 00 0a 14 0b 77 80 77 ...]@.....w w

0020 f5 0c cd 72 00 50 53 95 ec 85 ba 1b a1 87 50 18 ...r PS.....P

0030 00 ff 8d 18 00 00 47 45 54 20 2f 77 69 72 65 73GE T /wires

0040 68 61 72 6b 2d 6c 61 62 73 2f 48 54 54 50 2d 65 hark-lab s/HTTP-e

0050 74 68 65 72 65 61 6c 2d 6c 61 62 2d 66 69 6c 65 thereal- lab-file

0060 33 2e 68 74 6d 6c 20 48 54 54 50 2f 31 2e 31 0d 3.html H TTP/1.1

0070 0a 48 6f 73 74 3a 20 67 61 69 61 2e 63 73 2e 75 .Host: g aia.cs.u

0080 6d 61 73 73 2e 65 64 75 0d 0a 43 6f 6e 6e 65 63 mass.edu ..Connec

0090 74 69 6f 6e 3a 20 6b 65 65 70 2d 61 6c 69 76 65 tion: ke ep-alive

00a0 0d 0a 55 70 67 72 61 64 65 2d 49 6e 73 65 63 75 ..Upgrad e-Insecu

00b0 72 65 2d 52 65 71 75 65 73 74 73 3a 20 31 0d 0a re-Regue sts: 1..

00c0 55 73 65 72 2d 41 67 65 6e 74 3a 20 4d 6f 7a 69 User-Age nt: Mozi

00d0 6c 6c 61 2f 35 2e 30 20 28 57 69 6e 64 6f 77 73 lla/5.0 (Windows

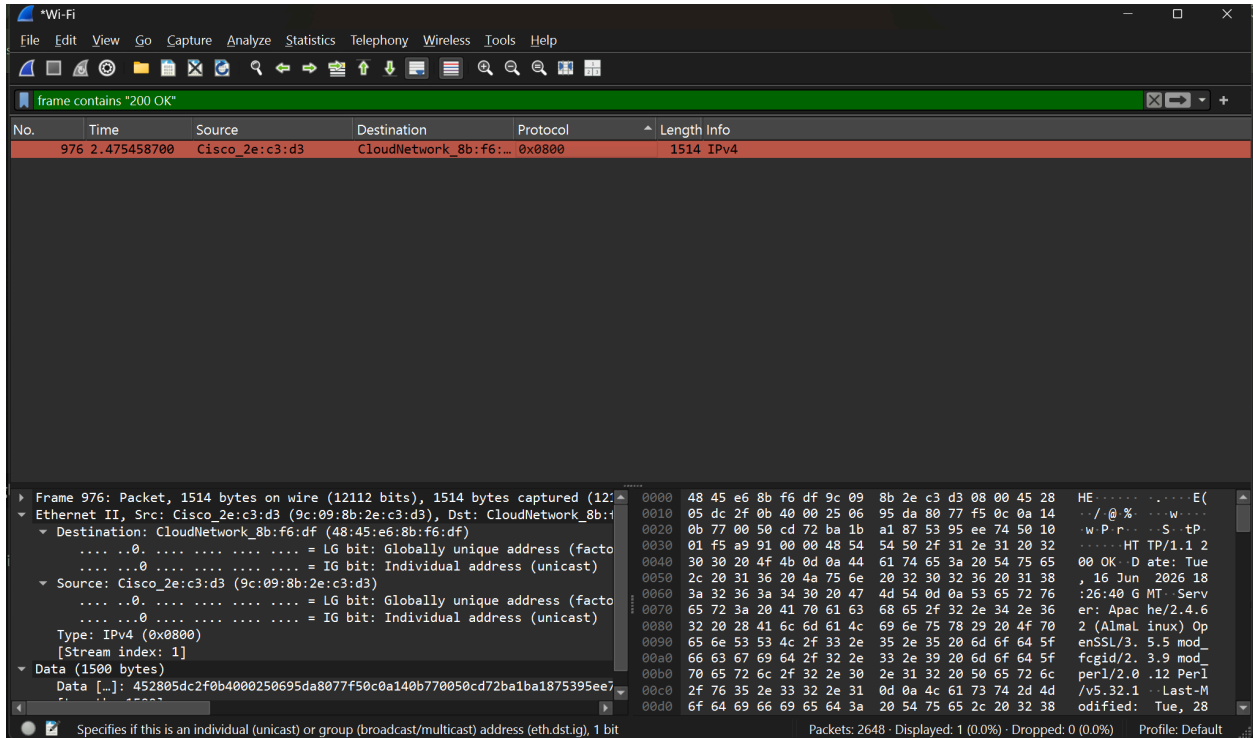
Ethernet (eth), 14 bytes

Packets: 2648 - Dropped: 0 (0.0%)

Profile: Default

1. My computer's MAC address (Source): **48:45:e6:8b:f6:df**
2. Destination MAC address: **9c:09:8b:2e:c3:d3**
This is a Cisco device (your router/gateway), NOT gaia.cs.umass.edu. The Ethernet frame only travels to your local router. Your router then handles getting the packet the rest of the way to gaia.cs.umass.edu using IP routing.
3. Frame type field: **0x0800 = IPv4**
4. The Ethernet header is always 14 bytes:
 - a. 6 bytes destination MAC
 - b. 6 bytes source MAC
 - c. 2 bytes type fieldThen after that comes IP header (20 bytes) + TCP header (20 bytes) = 40 more bytes, then the HTTP payload starts.
So: 14 + 20 + 20 = 54 bytes, meaning "G" appears at byte 55 (counting from 1).

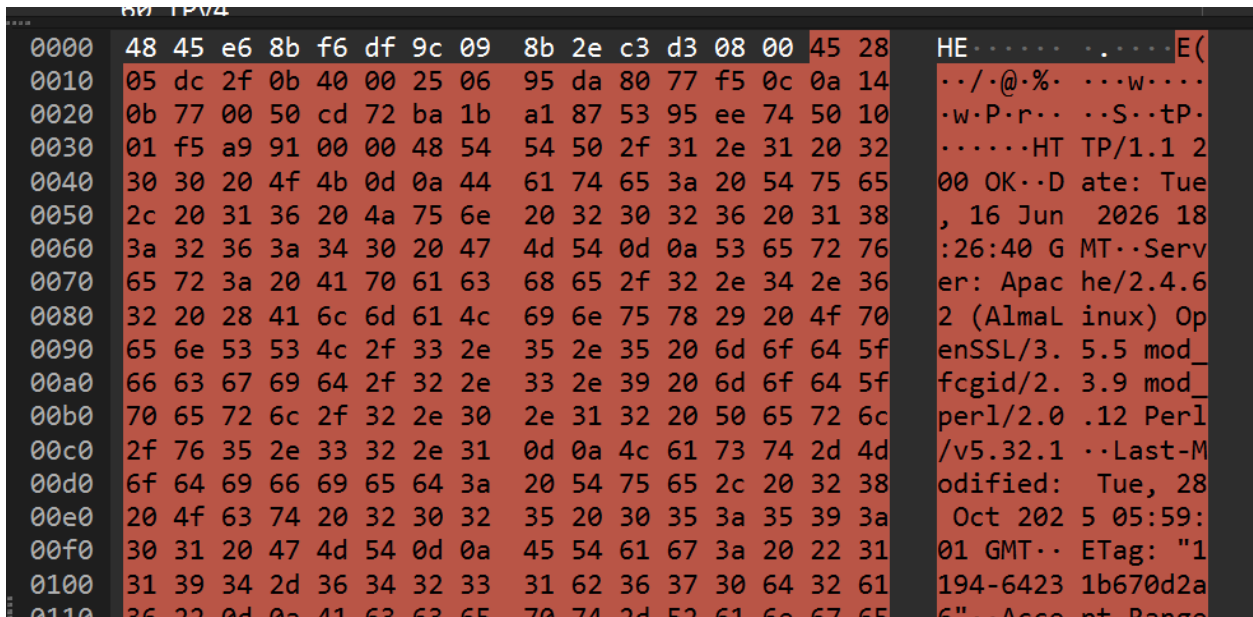
Ethernet and ARP Lab



- Source MAC address (Ethernet source of response): **9c:09:8b:2e:c3:d3**
(Cisco_2e:c3:d3)

This is your router/gateway again, NOT gaia.cs.umass.edu directly, for the same reason as before. The response comes from gaia, but your router is the one delivering it to you on the local network.

- Destination MAC address: 48:45:e6:8b:f6:df (CloudNetwork_8b:f6:df)
Yes, this **is** my computer's MAC address.
- Frame type field: **0x0800 = IPv4**, Same as the GET packet.



- byte 61, he "O" (4f) is at offset 0x003d = 61

Ethernet and ARP Lab

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\bhavd>arp -a

Interface: 10.20.11.119 --- 0xb
Internet Address      Physical Address      Type
10.20.0.1             9c-09-8b-2e-c3-d3    dynamic
10.20.8.45            e8-aa-cb-e1-db-56    dynamic
10.20.8.64            e8-aa-cb-e1-c7-f0    dynamic
10.20.8.158           72-90-ca-e4-ab-dc    dynamic
10.20.8.166           76-cd-6a-4c-4f-58    dynamic
10.20.8.169           14-7f-67-e7-10-66    dynamic
10.20.10.176          14-7f-67-e7-10-66    dynamic
10.20.11.50           52-9b-ac-af-60-c2    dynamic
10.20.12.59           d0-3c-1f-3f-4f-d0    dynamic
10.20.12.83           66-94-11-c5-19-5b    dynamic
10.20.12.157          52-14-24-aa-75-df    dynamic
10.20.12.173          5e-e7-9c-ae-33-be    dynamic
10.20.12.182          18-cc-18-7c-2c-d9    dynamic
10.20.12.213          ae-df-3b-5f-af-1e    dynamic
10.20.12.233          14-7f-67-e7-10-66    dynamic
10.20.13.9            c6-a8-85-c3-8c-18    dynamic
10.20.13.16           d2-9a-a8-ba-63-df    dynamic
10.20.13.34           ac-19-8e-f3-fa-e3    dynamic
10.20.255.255         ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
224.32.32.51         01-00-5e-20-20-33    static
```

9. Interface: 10.20.11.119 --- 0xb

Internet Address	Physical Address	Type
10.20.0.1	9c-09-8b-2e-c3-d3	dynamic
224.0.0.22	01-00-5e-00-00-16	static
224.32.32.51	01-00-5e-20-20-33	static

Meaning of each column:

- Internet Address — the IP address of a device on the network that your computer has recently communicated with or needs to know about
- Physical Address — the MAC address corresponding to that IP address, learned through ARP
- Type: dynamic — this entry was learned automatically by the ARP protocol when your computer needed to communicate with that device. It expires and gets deleted after a timeout period
- Type: static — this entry was permanently set, either manually or by the operating system. The 224.x.x.x addresses are multicast addresses that Windows adds automatically and they don't expire

Most notable entry: 10.20.0.1 → 9c-09-8b-2e-c3-d3 is your default gateway/router — the same Cisco device MAC address we saw in the Wireshark capture!

Ethernet and ARP Lab

```
Administrator: Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PowerShellLatest

Loading personal and system profiles took 528ms.
PS C:\Users\bhavd> arp -d *
PS C:\Users\bhavd> arp -a

Interface: 10.20.11.119 --- 0xb
Internet Address      Physical Address      Type
10.20.0.1              9c-09-8b-2e-c3-d3    dynamic
224.0.0.22             01-00-5e-00-00-16    static
224.32.32.51           01-00-5e-20-20-33    static
224.32.32.166          01-00-5e-20-20-a6    static
224.32.32.167          01-00-5e-20-20-a7    static
PS C:\Users\bhavd>
```

ethernet-ethereal-trace-1.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

arp

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	AmbitMicrosy_a9:3d:...	Broadcast	ARP	42	Who has 192.168.1.1? Tell 192.168.1.105
2	0.001018	LinksysGroup_da:af:...	AmbitMicrosy_a9:3d:...	ARP	60	192.168.1.1 is at 00:06:25:da:af:73
6	13.542974	CnetTechnolo_73:8d:...	Broadcast	ARP	60	Who has 192.168.1.117? Tell 192.168.1.104

Frame 6: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0

Ethernet II, Src: CnetTechnolo_73:8d:ce (00:80:ad:73:8d:ce), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

Destination: Broadcast (ff:ff:ff:ff:ff:ff)

...1... = LG bit: Locally administered address (this address only)

...1... = IG bit: Group address (multicast/broadcast address only)

Source: CnetTechnolo_73:8d:ce (00:80:ad:73:8d:ce)

...0... = LG bit: Globally unique address (factory default)

...0... = IG bit: Individual address (unicast)

Type: ARP (0x0806)

[Stream index: 2]

Padding: 00000000000000000000000000000000

Address Resolution Protocol (request)

Address Resolution Protocol: Protocol

Packets: 17 - Displayed: 3 (17.6%)

Profile: Default

10. Source and destination addresses in the ARP request:
 - a. Source: 00:d0:59:a9:3d:68 (AmbitMicrosy)
 - b. Destination: ff:ff:ff:ff:ff:ff (Broadcast)
11. Frame type field: **0x0806 = ARP**
12.
 - a. Opcode begins at byte 21 from start of frame
 - b. Opcode value = 1 (request)
 - c. Yes, The sender's IP address 192.168.1.105 is included in the ARP message
 - d. The "question" appears in the Target IP address field (192.168.1.1) -that's the IP whose MAC is being asked for. The Target MAC is all zeros because that's what we're trying to find out.
13.
 - a. Opcode begins at byte 21 from start of frame (same as the request)
 - b. Opcode value = 2 (reply)
 - c. The "answer" appears in the Sender IP address and Sender MAC address fields of the reply:
 - i. 192.168.1.1 is at 00:06:25:da:af:73
 - ii. The router is saying "that IP you were asking about? It's me, and here's my MAC"
14. Source and destination addresses in the ARP reply:
 - a. Source: 00:06:25:da:af:73 (LinksysGroup — the router)
 - b. Destination: 00:d0:59:a9:3d:68 (AmbitMicrosy — the computer that asked)
 - c. this is unicast not broadcast, because the router knows exactly who to reply to
15. Packet 6 is an ARP request from 192.168.1.104 asking "Who has 192.168.1.117?"

There is no reply visible because:

 - a. The reply was sent directly/unicast to 192.168.1.104
 - b. The Wireshark computer (192.168.1.105) is a different machine
 - c. So that unicast reply was never seen by Wireshark